A MACROMODEL FOR A CMOS TRANSMISSION GATE

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Abstract

A macromodel to simulate the output waveform of a CMOS transmission gate circuit is developed using a linear regression model which includes the coefficients for the input voltage, output voltage, and the interaction terms between the input and output voltages with load capacitor and gate transistor size.

Introduction

VLSI circuits fabricated using CMOS technology are used in many applications. Because of the large number of components in a VLSI circuit chip, the simulation of the circuit by computer requires large amounts of computer storage and CPU time. In order to reduce the expense of simulation, circuit macromodels have been developed[1-6].

Recently a number of techniques have been developed for the simulation of MOS circuits. These new techniques can be divided into four general categories: those which incorporate IC macromodels into a circuit simulation program[1], those which use a highly simplified device-level model in conjunction with relaxed circuit simulation schemes[2], those which use huge data sets curve-fitted into functions or look-up tables[3-5], and those which use analytic equations[6]. In this paper we develop a macromodel which attempts to simulate the output waveform of a transmission gate.

The Macromodel

The transmission gate can be modeled as a discrete system described by

$$H(Z) = \frac{V_o(z)}{V_{in}(z)} = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots}$$

where

$V_{in}(z)$ = Z transform of $V_{in}(t)$, the input voltage, and

$V_o(z)$ = Z transform of $V_o(t)$, the output voltage,[7].

By rearranging the above equation, we can write

$$V_o(z) = \{(a_1 z^{-1} + a_2 z^{-2} + \ldots)V_o(z) - (b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots)V_{in}(z)\}/a_0.$$

The Z-domain equation above can be transformed into the discrete time domain equation below:

$$V_o(t_n) = c_0 + c_1 \cdot V_{in}(t_n) + c_2 \cdot V_{in}(t_{n-1}) + c_3 \cdot V_{in}(t_{n-2}) + c_4 \cdot V_o(t_{n-1}) + c_5 \cdot V_o(t_{n-2}) + e_n, \text{ for } t_0 \leq t_n \leq t_N,$$

where $V_{in}(t_n)$ and $V_o(t_n)$ are the input and output voltages at t_n, and ε_n is the error term. The coefficients $c_0, c_1, \ldots, c_5$ are dependent on the output load capacitance, C_L, and the width-to-length ratio of the gate transistor, S. Hence, $c_i = f_i(C_L,S)$, where $i=0,1,\ldots,5$, and the model becomes:

$$V_o(t_n) = f_0(C_L,S) + f_1(C_L,S)V_{in}(t_n) + f_2(C_L,S)V_{in}(t_{n-1}) + f_3(C_L,S)V_{in}(t_{n-2}) + f_4(C_L,S)V_o(t_{n-1}) + f_5(C_L,S)V_o(t_{n-2}) + e_n, \quad \text{for } t_0 \leq t_n \leq t_N.$$

The function $f_i(C_L,S)$ can be any function that includes C_L and S. For example: $f_i(C_L,S) = K_{0i} + K_{1i}*S + K_{2i}*C_L + K_{3i}*S*C_L + K_{4i}*S/C_L + K_{5i}*C_L/S + K_{6i}*\log(S/C_L)$, where K_{ji} for $j=0,1,2,\ldots$ are constants. A correlation analysis is used to discard the low correlation coefficient terms in the expression for $f_i(C_L,S)$. The remaining K_{ji} Coefficients in the $f_i(C_L,S)$ are then obtained by using the time series regression method[8] to minimize the error in the model equation.

The model can be expanded to include other parameters, such as transistor threshold voltage, and control gate input voltage.

Although the macromodel does not include higher order terms of the input and output voltages, the preliminary results using a range of different input conditions indicate that the macromodel produces a voltage response which is very close to the voltage response obtained with the SPICE program.

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Simulation Results

Figure 1, is the schematic diagram of a CMOS transmission gate. In order to simplify the model, both of the n-channel and p-channel transistors in the transmission gate are assumed to have the same width to length ratio.

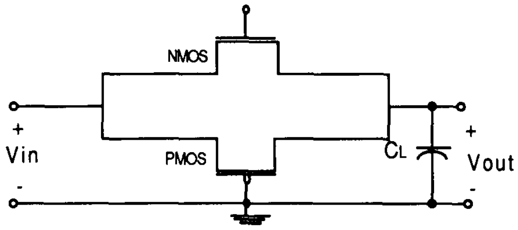

Figure 1. Schematic diagram of the CMOS transmission gate.

Three simulation examples have been computed for a transmission gate having a load capacitance of 1.0 pF. The channel width to length ratios used in these simulation are S = 15, 7.5, and 5.

The coefficient functions used for all three simulations are:

$f_0(C_L,S)=0.00016337-0.0000632*\log(S/C_L)$

$f_1(C_L,S)=0.21590228 + 0.41010529*C_L/S$
$+0.00226918*S/C_L-0.10821625*\log(S/C_L)$

$f_2(C_L,S)= 0.03170164 -0.03206234*\log(S/C_L)$
$+ 0.00393844*S/C_L$

$f_3(C_L,S) = 0.11848756 - 0.15846948*C_L/S$
$+ 0.08022315*\log(S/C_L)$

$f_4(C_L,S) = 2.07594846 - 0.01555093*S/C_L$
$- 0.0898809*\log(S/C_L)$

$f_5(C_L,S) = -1.12192035 +0.00816620*S/C_L$
$+0.13579712*\log(S/C_L)$

Figures 2,3, and 4, are plots of input voltage waveform (solid line), SPICE simulation output voltage waveform (dotted line), and macromodel simulation output voltage waveform (dashed line). These curves show that the macromodel results are very close to the SPICE results when $C_L/S < 0.1$. The smaller the ratio C_L/S is, the more closely the results correspond.

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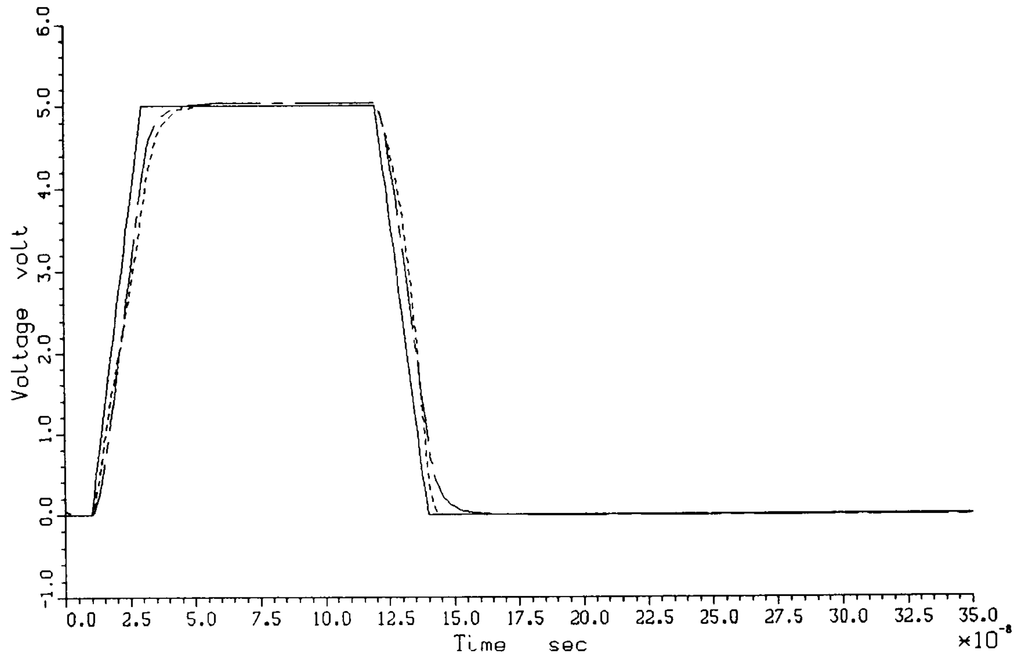

Figure 2. Output voltage for C_L = 1.0 pF and S = 15.

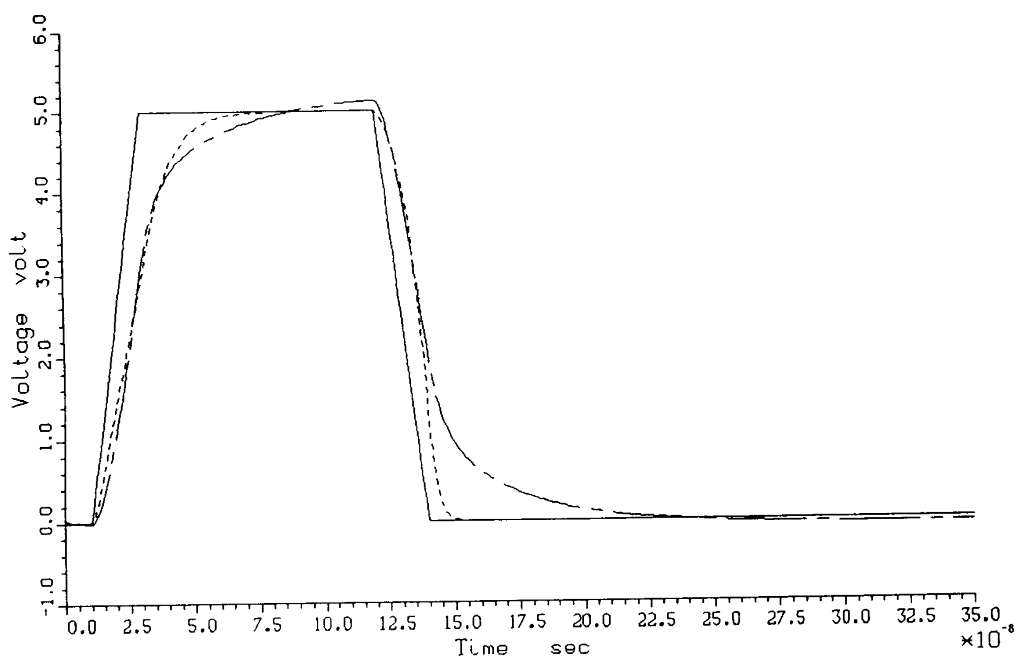

Figure 3. Output voltage for C_L = 1.0 pF and S = 7.5.

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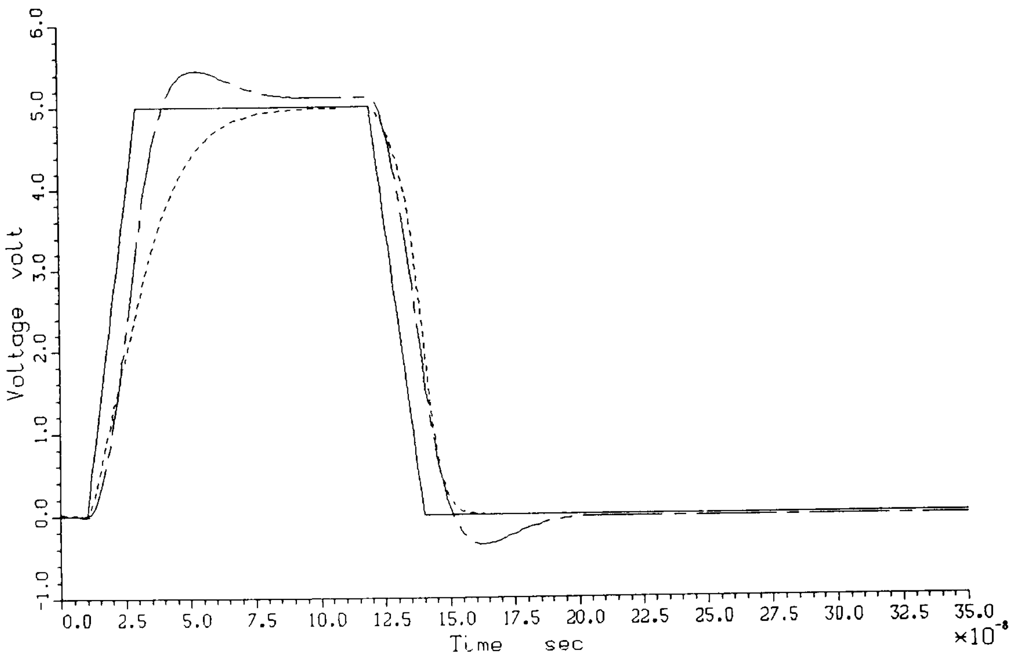

Figure 4. Output voltage for C_L = 1.0 pF and S = 5.

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